

Omada OCNA Cheat Sheet

Quick revision notes for both tracks: Wireless and Routing & Switching. Verify ports and menu names in the official Omada docs for your controller version.

1. Omada SDN basics

- **Components:** Omada Controller + EAP (access points) + switches + gateways (ER routers). One platform, centrally managed.
- **Controller types:** Hardware (OC200, OC300), Software (installed on PC/server), Cloud-Based (remote, no local controller hardware).
- **Standalone mode:** one device managed through its own web UI (small setups). **Controller mode:** central management, sites, provisioning, monitoring, portal.
- **Site:** logical group of devices with own settings (one branch/location). **Adoption:** controller takes over a device (shown as Pending until adopted). After adoption, config is provisioned.
- **Discovery:** same L2 network (UDP 29810), or Layer 3 via DHCP option 138 / DNS / manual 'Inform URL' on the device.

Port	Protocol	Purpose (Software Controller)
8088	TCP	HTTP management
8043	TCP	HTTPS management
8843	TCP	HTTPS portal
29810	UDP	Device discovery
29811	TCP	Manage (v1)
29812	TCP	Adopt (v1)
29813	TCP	Upgrade (v1)
29814	TCP	Manage (v2)

2. Wi-Fi standards

Standard	Name	Band	Notes
802.11b	-	2.4 GHz	11 Mbps
802.11a	-	5 GHz	54 Mbps
802.11g	-	2.4 GHz	54 Mbps
802.11n	Wi-Fi 4	2.4 / 5	MIMO, up to ~600 Mbps
802.11ac	Wi-Fi 5	5 GHz only	MU-MIMO (downlink), 80/160 MHz
802.11ax	Wi-Fi 6 (6E adds 6 GHz)	2.4 / 5 (6E: + 6)	OFDMA, UL/DL MU-MIMO, TWT, BSS coloring, 1024-QAM
802.11be	Wi-Fi 7	2.4 / 5 / 6	320 MHz channels, MLO, 4096-QAM

- **2.4 GHz:** longer range, better wall penetration, only 3 non-overlapping 20 MHz channels (**1, 6, 11**), crowded (microwaves, Bluetooth). **5 GHz:** more channels, faster, shorter range, DFS channels (radar). **6 GHz:** Wi-Fi 6E/7 only, cleanest, shortest range, WPA3 required.
- **Wider channel** = more throughput but fewer channels and more noise. Dense environments: use 20 MHz (2.4) / 20-40 MHz (5).
- **MIMO** = multiple spatial streams. **MU-MIMO** = several clients at the same time. **Beamforming** = focus signal at a client. **OFDMA** = split channel into resource units to serve many clients per transmission.
- **RSSI** = received signal strength (dBm, closer to 0 is stronger; -67 dBm is a common design target for good service). **SNR** = signal minus noise floor; higher is better.
- **PoE:** 802.3af = 15.4 W, 802.3at (PoE+) = 30 W, 802.3bt = 60 W / 90 W (PSE output).

3. Wireless security and roaming

Item	Key points
WEP / WPA (TKIP)	Obsolete, insecure.
WPA2	AES-CCMP. Personal (PSK) or Enterprise (802.1X).
WPA3	Personal uses SAE (protects against offline dictionary attacks, forward secrecy). Enterprise has stronger crypto. PMF mandatory. OWE = encrypted open networks.
802.1X	Supplicant (client) - Authenticator (AP/switch) - Authentication server (RADIUS). RADIUS ports 1812/1813.
802.11k	Neighbor reports: tells client which APs/channels are nearby (faster scanning).
802.11v	BSS Transition Management: AP suggests the client move to another AP/band (load balancing, steering).
802.11r	Fast BSS Transition: pre-shared key material, fast handoff (VoIP). Needs same SSID and security on all APs.
802.11w	Protected Management Frames (anti-deauth attacks).
802.11e / WMM	QoS: voice, video, best effort, background.
Band steering	Pushes dual-band clients to 5 GHz.
Mesh	Wireless uplink between APs; consumes airtime, so wired uplinks are better.

4. Portal and hotspot

- **Captive portal:** redirects new clients to a login/splash page before granting access. Typical authentication: simple password, local user, voucher, SMS, external RADIUS, external portal.
- **Voucher:** generated codes limited by time/data; ideal for guest WiFi in hotels, cafes.
- **Pre-authentication access:** resources reachable before login (e.g. payment page). **Authentication-free:** devices allowed without portal.
- Put guest SSIDs on a **separate VLAN**, apply ACLs/rate limits, enable client isolation.

5. Demand analysis, planning and optimization

- **Demand analysis:** number of clients, concurrency (% active), apps and bandwidth per client, coverage area, building materials, device types (2.4 / 5 GHz capable), roaming needs.
- **Capacity formula:** clients x concurrency x bandwidth per client. Example: 100 x 50% x 2 Mbps = 100 Mbps.
- **Site survey:** predictive (plan/map) and on-site (measure RSSI, noise, interference) to choose AP count and placement.
- **Optimization:** avoid co-channel interference (neighbors on the same channel), use balanced / lower transmit power (small cells), keep slight overlap between cells for roaming, prefer non-DFS channels where stability matters, disable low data rates in dense areas, wired uplinks, enough PoE budget.

6. Network fundamentals

Layer	Name	Examples / PDU
7	Application	HTTP, DNS, DHCP, SNMP
6	Presentation	TLS, encoding
5	Session	Sessions
4	Transport	TCP, UDP (segment / datagram)
3	Network	IP, ICMP, routers (packet)
2	Data Link	Ethernet, MAC, switches, VLAN (frame)
1	Physical	Cables, hubs, radio (bits)

Prefix	Mask	Usable hosts	Block size
/24	255.255.255.0	254	256
/25	255.255.255.128	126	128
/26	255.255.255.192	62	64
/27	255.255.255.224	30	32
/28	255.255.255.240	14	16
/29	255.255.255.248	6	8
/30	255.255.255.252	2	4

- **Private IPv4:** 10.0.0.0/8, 172.16.0.0/12 (172.16-31.x.x), 192.168.0.0/16. **APIPA:** 169.254.0.0/16 = no DHCP answer. **Loopback:** 127.0.0.0/8.
- **Usable hosts** = 2^(host bits) - 2. **Network address** = lowest address in the block, **broadcast** = highest.
- **DHCP (DORA):** Discover, Offer, Request, Acknowledge. **ARP:** IP to MAC. **NAT:** private to public. **DNS:** name to IP. TCP = connection-oriented, UDP = connectionless.

Port	Service	Port	Service
20/21	FTP	123	NTP (UDP)
22	SSH	161/162	SNMP
23	Telnet	389	LDAP
25	SMTP	443	HTTPS
53	DNS	514	Syslog (UDP)
67/68	DHCP server/client	1812/1813	RADIUS
69	TFTP	3389	RDP
80	HTTP		

7. Switching

- **Switch** learns source MACs, forwards by destination MAC; unknown/broadcast = flood. Each VLAN = one broadcast domain.
- **VLAN (802.1Q):** 4-byte tag, IDs 1-4094. **Access port** = one VLAN, untagged. **Trunk port** = multiple VLANs, tagged; **native VLAN / PVID** = untagged traffic on trunk.
- **STP (802.1D)** prevents loops; slow convergence. **RSTP (802.1w)** fast convergence. Root bridge = lowest bridge ID (priority, then MAC; default priority 32768). RSTP roles: root, designated, alternate, backup. States: discarding, learning, forwarding.
- **LAG / LACP (802.3ad, 802.1AX):** bundle links for bandwidth + redundancy; settings must match on both ends.
- **Port mirroring:** copy traffic to analyzer port. **IGMP snooping:** limit multicast to interested ports. **Storm control:** limit broadcast/multicast/unknown unicast. **Loopback detection:** blocks self-looped ports. **DHCP snooping:** blocks rogue DHCP servers (trusted ports only). **Port security / MAC binding:** limit which MACs may use a port.
- **QoS:** prioritize voice/video. **Rate limit / bandwidth control:** cap per client or port.

8. Routing, gateway, VPN and security

- **Routing table:** longest prefix match wins. **Default route** 0.0.0.0/0 = anything not matched. Next hop of a static route must be on a directly connected network.
- **Static** routes = manual, simple. **Dynamic** (e.g. OSPF, link-state IGP) = learned automatically, adapts to changes.
- **Inter-VLAN routing** needs a Layer 3 device: router (router-on-a-stick) or L3 switch.
- **Multi-WAN:** *load balancing* uses several WANs at once; *link backup* = failover to a second WAN. **Policy routing:** choose WAN/next hop by source, destination or service.
- **Port forwarding / virtual server:** external port to internal host. **DMZ:** isolated segment for public servers. **DHCP reservation:** fixed IP per MAC. **DHCP relay:** forwards DHCP to a server in another network.
- **ACL:** permit/deny by source, destination, protocol, port; evaluated top to bottom (first match). Omada has Gateway, Switch and EAP ACLs.
- **VPN:** site-to-site (connect two networks, IPsec) vs client-to-site (remote user). Other: OpenVPN, L2TP, WireGuard (depending on model/firmware).

9. Troubleshooting

- **Method:** define the problem, check the physical layer first (power, PoE, cable, link LEDs), then IP settings, then services, then configuration. Change one thing at a time.
- **Windows tools:** `ipconfig /all` (IP, gateway, DNS), `ipconfig /release /renew`, `ping` (ICMP reachability), `tracert` (path hop by hop), `nslookup` (DNS), `arp -a` (ARP table), `netstat`.

Symptom	Likely cause
169.254.x.x address	No DHCP response (DHCP server, VLAN, relay, wireless auth).
Ping to IP works, names fail	DNS problem.
Can reach local hosts, not internet	Wrong or missing default gateway, NAT, WAN issue.
Poor Wi-Fi, low speed, drops	Co-channel interference, too-high power, wide channels, weak RSSI, too many clients per AP.
Roaming issues, sticky clients	Power too high, no 802.11k/v/r, different security on APs, too little or too much overlap.
AP shows Disconnected	Power/PoE, cable, no IP route to controller, firewall blocking controller ports.
Network slows, many broadcasts	Layer 2 loop (check STP / loopback detection), storm control.
Cannot connect after enabling 802.11r/WPA3	Old client does not support it: test with feature disabled.

10. Exam tips

- Pass mark is reported as **70%**; the exam is taken online in about an hour. Read each question for the **best** answer, and eliminate wrong options first.
- Know the **differences:** access vs trunk, STP vs RSTP, load balancing vs link backup, 802.11k vs v vs r, standalone vs controller mode, Site vs Controller.
- Practice quick subnetting (block size method) and the capacity formula. Memorize the Omada port list and PoE standards.